

# Vibe-IC — ALL Steps, by Stage, in Order (v2.2.0)

**Plugin 0.2.6.** ONE list. Every step of the flow, grouped by stage, in execution order, with a single continuous number (1 → 33) that STARTS at Stage 1's first step (Spec-to-RTL), matching the runner / flow-YAML source of truth (step 1 = Spec-to-RTL; Phase 1 doc generation is the lettered pre-flow step D1, broken out here as D1–D5). Two parallel tracks (Analog, Mixed-signal) are listed after the main flow. Source of truth = the runners.

The runner's *internal implementation markers* (Phase 1 [1/15]... [15/15] + 19 sub-steps; Phase 2/3 def step\_\*) are a finer-grained code view, auto-generated in **FLOW\_STEPS\_GENERATED.md** by `flow_doc_emit.py`. They map onto the stages below; this doc is the single human-facing ordered list.

Stage map: **0** Spec/Docs (Phase 1) · **1** RTL+Verify · **2** Synthesis+DFT · **3** Physical+Sign-off (Phase 3) · **4** Output+Tapeout. (Phases: Phase 1 = Stage 0; Phase 2 ≈ Stage 1-2; Phase 3 ≈ Stage 3-4.)

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## Phase 1 (pre-flow) — Spec & Documents · lettered D1–D5 (not part of the 1→33 counter)

| #  | Step                                                          | Tool / How                                |
|----|---------------------------------------------------------------|-------------------------------------------|
| D1 | Ingest & text extraction (prompt or vendor docs → input_doc/) | phase1_doc_one_shot_runner ([1/15])       |
| D2 | Generate L1–L13 core design-layer docs                        | deterministic extractors ([2/15]–[14/15]) |
| D3 |                                                               | [14b/15]–[14d/15] overlays                |

| #  | Step                                                     | Tool / How                                            |
|----|----------------------------------------------------------|-------------------------------------------------------|
|    | Generate L14–L23<br>protocol / timing /<br>skeleton docs |                                                       |
| D4 | Protocol-class<br>synthesis dispatch<br>(81 classes)     | [14e/15]–[14e3/15]<br>(is_<proto> +<br><proto>_synth) |
| D5 | Coverage / parity<br>report                              | [15/15]                                               |

## Stage 1 — RTL Generation & Verification (Phase 2, front)

| # | Step                                    | Tool / How                                                                       |
|---|-----------------------------------------|----------------------------------------------------------------------------------|
| 1 | Spec-to-RTL (author<br>RTL from L-docs) | spec-to-rtl skill<br>(runner WAIVES<br>step_rtl_gen for<br>rtl_gen=null classes) |
| 2 | Lint                                    | eda_lint + hygiene<br>gates                                                      |
| 3 | CDC / RDC check                         | cdc-check                                                                        |
| 4 | Simulation                              | testbench-gen +<br>eda_simulate +<br>coverage                                    |
| 5 | Formal verification                     | formal-verify +<br>assertion-gen<br>(informational<br>waiver if no model)        |
| 6 | FPGA early prototype                    | eda_fpga_compile /<br>eda_fpga_program +<br>on-board BIST → .sof                 |

## Stage 2 — Synthesis & DFT (Phase 2, back)

| # | Step             | Tool / How                               |
|---|------------------|------------------------------------------|
| 7 | Constraint setup | constraint-gen →<br>*.sdc + 3-corner PVT |

| #  | Step                        | Tool / How                                 |
|----|-----------------------------|--------------------------------------------|
| 8  | SDC validation              | SDC lint                                   |
| 9  | Synthesis (Yosys)           | eda_synth + synth-doctor (+ tie-cell pass) |
| 10 | Pre-layout STA              | eda_sta_mcorner (SS/TT/FF)                 |
| 11 | DFT insertion (scan + ATPG) | dft-insert + atpg + eda_dft                |
| 12 | Post-DFT optimization       | resynth / buffering                        |
| 13 | Equivalence check (LEC)     | equivalence-check + Yosys equiv            |

## Stage 3 — Physical Design & Sign-off (Phase 3)

| #  | Step                          | Tool / How                                                                     | Open-source status                |
|----|-------------------------------|--------------------------------------------------------------------------------|-----------------------------------|
| 14 | Floorplan + PDN               | eda_pnr (init)                                                                 | ✓                                 |
| 15 | Clock planning                | clock-planning skill                                                           | ✓                                 |
| 16 | Placement (global + detailed) | eda_pnr                                                                        | ✓                                 |
| 17 | CTS                           | eda_pnr enable_cts=true                                                        | ✓                                 |
| 18 | Post-CTS hold fixing          | repair_timing -hold                                                            | ✓                                 |
| 19 | Routing (global + detailed)   | eda_pnr<br>enable_detailed_route=true                                          | ✓                                 |
| 20 | Parasitic extraction → SPEF   | OpenRCX extract_parasitics - ext_model_file<br>rules.openrcx.sky130A.nom.magic | ✓ FIXED v0.2.5 (real 268 KB SPEF) |
| 21 |                               | eda_sta_mcorner                                                                | ✓                                 |

| #  | Step                                                 | Tool / How                                                                       | Open-source status                         |
|----|------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------|
|    | Post-route STA (MMMC)                                |                                                                                  |                                            |
| 22 | IR drop                                              | OpenROAD PSM<br>analyze_power_grid                                               | ✓ FIXED<br>v0.2.4                          |
| 23 | EM check                                             | PSM -enable_em                                                                   | ✓ FIXED<br>v0.2.4                          |
| 24 | Antenna check                                        | OpenROAD check_antennas                                                          | ✓ FIXED<br>v0.2.4                          |
| 25 | Signal integrity (crosstalk)                         | SPEF coupling-cap screen (Cc/ (Cc+Cg))                                           | ✓ WIRED<br>v0.2.6<br>(advisory)            |
| 26 | Post-layout gate-level sim (+SDF)                    | eda_simulate                                                                     | ✓                                          |
| 27 | Physical verification (DRC / LVS / ERC + PERC-equiv) | KLayout DRC + LVS sign-off chain (below) + Magic ERC + perc_equivalent aggregate | ✓ DRC/ LVS/ERC ;<br>PERC-equiv (~70% auto) |
| 28 | ECO repair loop                                      | eco-plan                                                                         | ✓                                          |

**Step 27 — LVS sign-off chain** (under step\_lvs): (1) structural LEC `eda_lvs yosys_equiv` → (2) device-level `eda_extraction + netgen` → (3) powered-netlist `write_verilog -include_pwr_gnd` → (4) port labels `magic_port_extract_emit (Route A) / lvs_def_port_seed (Route B)` → (5) **mandatory** `lvs_signoff_guard` (RAISES on a portless/vacuous match).

**Step 27 — PERC-equivalent coverage** (`perc_equivalent.{rpt,json}` + `PERC_SIGNOFF_MEM0.md`, v0.2.7-2.8): open-source stand-in for commercial Calibre PERC. AUTOMATED: antenna / IR / EM / floating-nets (verdicts read from steps 22-25/ERC). GUARDBAND: EM current-density (<0.5 mA/μm) + ≥2×2 vias. MANUAL\_REVIEW (never auto-PASS, pending checklist): latch-up spacing/device-physics, cross-voltage-domain — auto-N/A for core-only macros / single-supply designs. **Latch-up well-tap presence** (v0.2.10) is now AUTOMATED: a routed DEF with 0 valid well/substrate-tap cells is a conclusive `WELLTAP_GAP FAIL` → `PERC_EQUIV_FAIL` (catches the real v0.1.45-class tapcell-skip silicon bug;

the real spm/subservient/neorv32 routed DEFs all ship 0 taps). Only tap *presence* is automated — tap spacing + the device-physics latch-up criterion ( $V_{hold} > V_{dd}$ , SCR  $\beta$ -product, guard-ring efficacy) stay MANUAL (an adversarial panel showed DEF-only spatial density/max-distance over-claims). The **ESD presence** check (v0.2.8) recognises sky130 IO integral-clamp pads (gpiov2/hvc/lvc/clamped/ esd) — validated on the real Caravel chip\_io.def (flips a 612-filler ring from a false ESD\_MISSING to PRESENT, structural fillers excluded) with negation guards (unclamped $\neq$ ESD) + esd-before-structural ordering (gpiov2\_corner\_pad=ESD), and reports esd\_presence: PRESENT | MISSING | N/A. The **ESD discharge-path topology** check (v0.2.9, a NEW **AUTOMATED** category) automates the *connectivity half* of ESD sign-off from DEF COMPONENTS + NETS: domain-loop completeness (a supply clamp with no matching ground-return clamp = open loop), clamp stitching (each clamp tied to BOTH a power and a ground net), and rated-cell membership. A TOPOLOGY\_GAP is a conclusive automated FAIL  $\rightarrow$  PERC\_EQUIV\_FAIL; TOPOLOGY\_OK is necessary-but-not-sufficient. **This shrinks the ESD MANUAL\_REVIEW residual to exactly the device-physics half** — clamp HBM/CDM sizing (TLP/It<sup>2</sup>), inherited from the rated library-cell datasheet (an adversarial panel confirmed connectivity can never prove sizing, so the presence category stays MANUAL). Validated on the real Caravel chip\_io.def: 63 ESD-pad instances, all 3 domain loops closed, 0 dangling, 0 unrated  $\rightarrow$  TOPOLOGY\_OK. The **cross-voltage-domain** check (v0.2.11) now counts power domains robustly from NETS + SPECIALNETS USE POWER/GROUND + recognised supply-net families (fixing the real Caravel single-supply mis-count — its supplies are declared via NETS, not SPECIALNETS, so the old SPECIALNETS-only path wrongly returned single-supply). Conclusive automated FAIL:  **$\geq 2$  domains AND 0 level-shifter/isolation/IO-crossing cells**  $\rightarrow$  XDOMAIN\_GAP  $\rightarrow$  PERC\_EQUIV\_FAIL (an inter-domain signal is guaranteed un-shifted). When  $\geq 1$  crossing cell is present the category stays **MANUAL\_REVIEW** (an adversarial panel ruled “a crossing cell exists somewhere”  $\neq$  “every crossing is shifted”; per-crossing direction lo $\rightarrow$ hi/hi $\rightarrow$ lo + isolation-clamp efficacy are device physics). Conservative family-collapse keeps vccd1/vccd2 voltage splits DISTINCT (never merges a real domain away); unresolved partitions degrade to INCOMPLETE, never silent N/A.

## Stage 4 — Output & Tapeout (Phase 3, close)

| #  | Step                         | Tool / How                                               | Open-source status |
|----|------------------------------|----------------------------------------------------------|--------------------|
| 29 | Power analysis               | pre + post layout                                        | ✓                  |
| 30 | Metal fill<br>(density fill) | OpenROAD<br>filler_placement<br>→ filled.def             | ✓ FIXED<br>v0.2.4  |
| 31 | Tapeout<br>checklist         | tapeout-<br>checklist +<br>signoff_audit<br>(4/4 strict) | ✓                  |
| 32 | GDSII output                 | eda_gds +<br>def2gds (only if<br>step 27 clean)          | ✓                  |
| 33 | FPGA final<br>sign-off       | recompile + on-<br>board test +<br>attestation           | ✓                  |

## Parallel track — Analog A1–A9 (analog\_one\_shot\_runner.py, runs alongside Stages 1-3)

| #  | Step                                              | Output                                               |
|----|---------------------------------------------------|------------------------------------------------------|
| A1 | spec_extract                                      | analog/<block>/<br>A1_spec.json                      |
| A2 | topology_select                                   | A2_topology.json                                     |
| A3 | netlist_gen                                       | <block>.sp                                           |
| A4 | corner_sweep                                      | A4_corners.json                                      |
| A5 | layout (Magic)                                    | A5_layout.json                                       |
| A6 | per-block physical<br>verification (DRC +<br>LVS) | analog_a6_block_pv_check<br>(real per-block DRC+LVS) |
| A7 | post_layout_resim                                 | A7_postsim.json                                      |
| A8 | hardmacro_gen                                     | {.lef,.lib,.gds,.v} →<br>feeds Stage 3               |

| #  | Step                     | Output            |
|----|--------------------------|-------------------|
| A9 | hw_verify (HIL) / co-sim | A9_hw_verify.json |

**Parallel track — Mixed-signal M1–M4 (skill mixed-signal-cosim, no dedicated runner)**

| #  | Step                                           |
|----|------------------------------------------------|
| M1 | top merge (mixed_signal_m1_top_merge_check.py) |
| M2 | co-sim setup                                   |
| M3 | co-sim run                                     |
| M4 | integration verify                             |

**Totals**

**33 sequential steps** (Stage 1: 1-6 · Stage 2: 7-13 · Stage 3: 14-28 · Stage 4: 29-33), preceded by the lettered **Phase 1 pre-flow D1-D5 + 9 Analog (A1-A9) + 4 Mixed-signal (M1-M4)**, both parallel.

Pre-flight: P0 (mcp\_server\_health\_check, eda\_doctor).  
Orchestrator: vibe\_ic\_one\_shot\_runner.py runs Phase 1 → Phase 2 → Analog → Phase 3.

This is a **derived** list. The runner’s live implementation markers are auto-generated in FLOW\_STEPS\_GENERATED.md (flow\_doc\_emit.py --check fails CI on drift). 繁中版見 ALL\_STEPS\_v2.2.0.zh-TW.md.